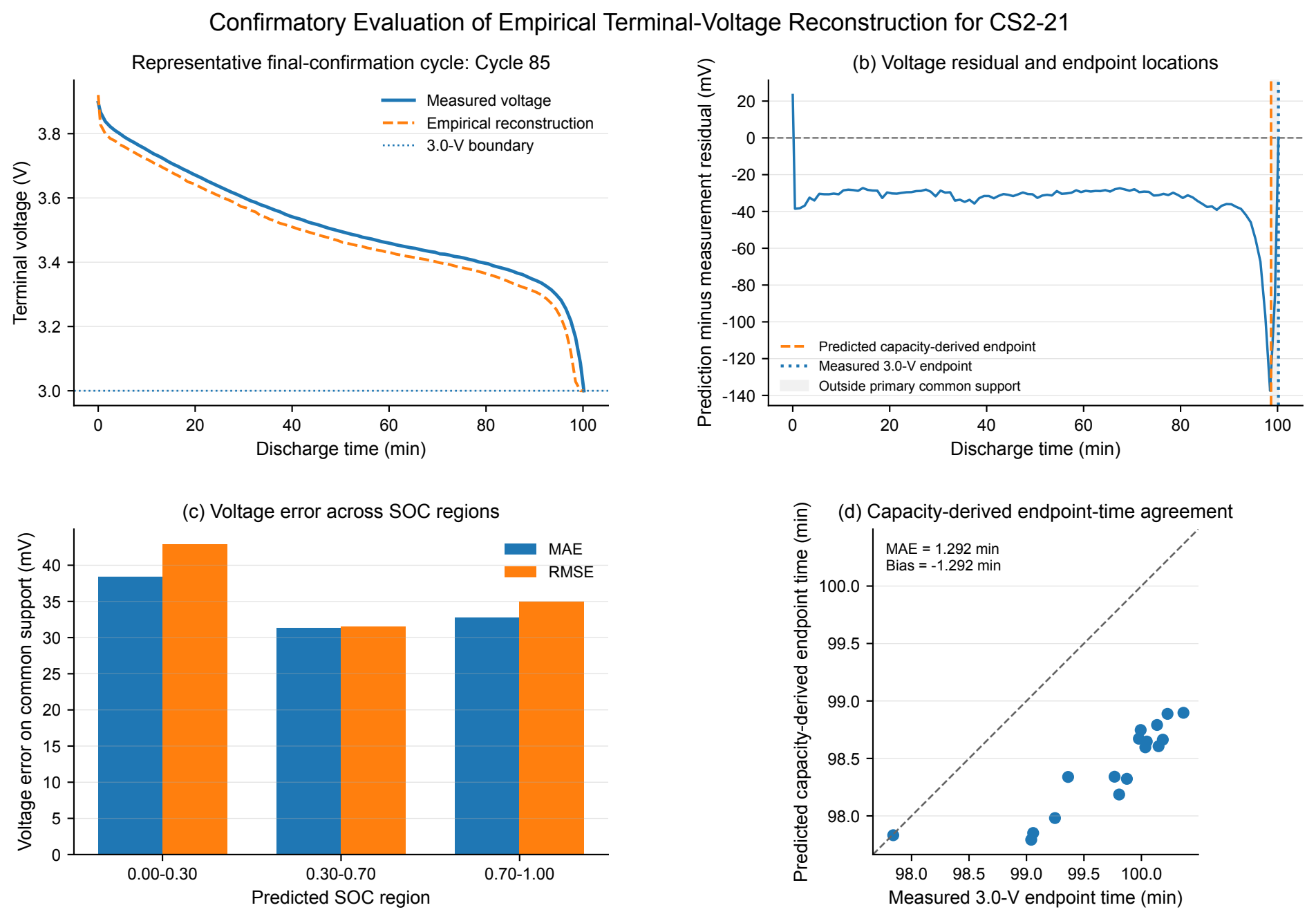
Confirmatory Evaluation of Empirical Terminal-Voltage Reconstruction for CS2-21
Representative final-confirmation cycle: Cycle 85
Measured voltage
Empirical reconstruction
3.0-V boundary
Terminal voltage (V)
Discharge time (min)
(b) Voltage residual and endpoint locations
Prediction minus measurement residual (mV)
Predicted capacity-derived endpoint
Measured 3.0-V endpoint
Outside primary common support
Discharge time (min)
(c) Voltage error across SOC regions
MAE
RMSE
Voltage error on common support (mV)
0.00-0.30
0.30-0.70
0.70-1.00
Predicted SOC region
(d) Capacity-derived endpoint-time agreement
MAE = 1.292 min
Bias = -1.292 min
Predicted capacity-derived endpoint time (min)
Measured 3.0-V endpoint time (min)